UNIVERSITÀ DEGLI STUDI FIRENZE

Scuola di Scienze Matematiche, Fisiche e Naturali

Laurea Magistrale in Data Science, Calcolo Scientifico & Intelligenza Artificiale

Master Thesis

ANTI-MONEY LAUNDERING WITH GRAPH NEURAL NETWORKS

NAME SURNAME

Advisor: *prof. Name Surname*

Co-Advisor: *prof. Name Surname*

Academic Year 2023-2024

ABSTRACT

CONTENTS

LIST OF FIGURES

"Citation"
— citation's author

1

INTRODUCTION

Money laundering (ML) is the criminal process of disguising or concealing the origin, identity, and ownership of illegally acquired funds. Its detection has become a foundational pillar of modern economic security. The rapid digitization of financial services has opened new possibilities for fraudulent activities. Undetected ML leads to massive economic losses for individuals and institutions, undermines the public trust in the financial system and frequently serves to finance criminal organizations. The scale of this phenomenon is staggering: the United Nations Office on Drugs and Crime (UNODC) estimates that between 2% and 5% of global GDP is laundered annually, with less than 1% of these illicit financial flows being successfully detected [12, 7]. According to the Canadian Anti Fraud Centre (CAFC), 704 million dollars were lost because of frauds in 2025 only, confirming the raising trend of the previous years [1]. Consequently, developing robust mechanisms to identify and stop it is critically important to preserving both market integrity and social stability.

1.1 TRADITIONAL AML METHODS

Anti Money Laundering is a critical component of the global financial system. Financial institutions ought to implement robust mechanisms to comply with regulatory standards and protect the integrity of the financial system [10].

Rule-based models have long been employed to detect suspicious transactions. These models consist of a set of predefined rules that are used to flag transactions that deviate from typical financial behavior. The criteria are typically based on regulatory requirements, historical data, and known patterns of illicit behavior. For example we could decide to flag transactions that exceed a certain amount, that rapidly transfer funds between accounts or that involve a high-risk jurisdiction.

The main advantage of this approach is its transparency and interpretability. Furthermore they are computationally inexpensive, easy to implement and they can operate in real-time [10]. Unfortunately there are also several limitations. They tend to produce a high amount of false positives, which leads to more work from human operators that will have to deal with the flagged transactions. They could be easily exploited by criminals who are aware of the rules and can tailor their illicit activities to avoid detection. This will be further discussed in Chapter 2.

Due to its tendency of producing a huge amount of false positives and the huge amount of data that nowadays is produced by financial institutions, the rule-based approach is becoming less and less effective.

At the same time big datasets allow the usage of data-driven solutions that can learn directly from historical data. First attempts in this direction involved standard machine learning approaches such as logistic regression and support vector machines [2]. Those models improved the state of the art but they still struggled to detect human brain-armed patterns due to their very limited parameter capacity [3].

For this reason the research field shifted toward deep learning solutions which offer a much wider and complex hypothesis space, allowing to capture high order relationships between data, at the cost of a much worse interpretability. Nowadays the challenge is to combine high performance together with explainability to allow the usage of effective models in the highly regulated financial field, a necessity increasingly emphasized by both financial institutions and supervisory authorities [8]. We will discuss this further in Chapter 3

1.2 THE GROWING CHALLENGE

Detecting fraud is becoming increasingly difficult. The sheer volume and velocity of digital transactions make manual oversight impossible and heavily strain classical machine learning architectures. Furthermore, the rise of decentralized finance (DeFi), Web3, and cryptocurrencies has introduced a paradigm of pseudo-anonymity that complicates traditional tracing.

On top of that, criminals are constantly developing more and more sophisticated methods to avoid being caught. As Cuèllar (2003) noted, *"raising the marginal cost of perpetrating criminal activity can itself conceivably have counterintuitive effects. To the extent that criminals engaged in crimes for profit are acting rationally, then raising the cost of crime can discourage smaller operations and clear the market allowing more sophisticated criminal* *organizations to innovate and take more control from less organized criminals"* [6]. Hence, to prevent inadvertently incentivizing sophisticated criminal networks, detection mechanisms must be updated continuously. The framework proposed in this study leverages deep learning models that support online updates. This ensures that the detection algorithm remains aligned with the most recent transaction patterns.

1.3 GNN FOR FRAUD DETECTION

The necessity of capturing complex interactions between accounts and transactions led to the idea of modeling the financial feed as a graph, where accounts are nodes and transactions are edges [11].

Definition 1.1 (Financial Graph)**.** *A financial graph is a directed graph* $G = (V, E)$*, where* V *represents a set of financial entities (e.g. bank accounts or individuals), and* E *represents a set of transactions between these entities. Each edge* $e \in E$ *is directed from the sender to the receiver and may be associated with features.*

GNNs rely on message passing to aggregate information from neighbors and learn representations of the nodes that will be given as input to a classifier or regressor to predict the probability that an account (Node-Level detection), a transaction (Edge-Level detection) or a subgraph (Graph-Level detection) is fraudulent. These methods allow to capture complex relationships, learn features automatically, handle dynamic graphs, and integrate multimodal data.

Formally the message passing process at the k-th layer in a GNN is defined as follows:

$$\begin{aligned} m_i^{(k)} &= \text{AGGREGATE}\left\{\text{MSG}\left(h_j^{(k-1)}, e_{ji}\right) : j \in \mathcal{N}(i)\right\} \\ h_i^{(k)} &= \text{UPDATE}\left(h_i^{(k-1)}, m_i^{(k)}, x_i\right) \end{aligned} \tag{1.1}$$

where $\mathcal{N}(i)$ denotes the set of neighboring nodes of node i, e_{ji} is the feature vector of the edge from node j to node i and x_i is the feature vector of node i. Following K iterations of message passing, the resultant node embeddings $H^{(K)}$ are then used to execute the designated task, in our case the detection of fraudulent accounts [4].

1.4 STRUCTURE OF THE DISSERTATION

2

FINANCIAL CRIME AND MONEY LAUNDERING

2.1 THE LIMITATIONS OF THE LONELY SCAMMER

Individual scammers typically rely on simple and intuitive techniques to introduce illicit funds into the legitimate economy. One of the most prevalent methods is known as "smurfing". In simple terms, smurfing involves breaking down a large sum of illicit money into many smaller, inconspicuous chunks to deposit them without raising suspicion.

More formally, individual scammers rely on smurfing to avoid Anti-Money Laundering (AML) detection mechanisms, which usually flag transactions above a specific threshold $\tilde{x}$. By splitting their initial illicit funds x into smaller transactions x_n, such that each transaction satisfies $x_n \leqslant \tilde{x}$, they minimize the risk of detection [5]. The mathematical expectation of profit π_s for a scammer engaging in smurfing can be defined as:

$$\pi_s = x(1 - \nu) \tag{2.1}$$

where x represents the total initial illicit funds, and $\nu \in (0, 1)$ is the friction cost or commission proportion associated with the smurfing process (e.g., the fee paid to mules).

While smurfing is effective for small amounts, its scalability is severely limited. When attempting to launder much larger volumes of funds, an individual scammer operating from a single account faces a significantly higher probability of detection $(1 - p)$, where p is the probability of successfully avoiding detection. If caught, the scammer is subjected to a severe monetary penalty T, which includes both the total confiscation of the funds and associated judicial fines [5]. Therefore, operating as a "lonely scammer" becomes mathematically and practically unviable for large-scale operations due to the massive expected cost of $(1 - p)T$.

2.2 FRAUD RINGS

In regard to overcome those issues, criminals have improved their techniques by employing highly sophisticated obfuscation tactics. Instead of acting alone, they form complex "fraud rings" to distribute illicit funds across vast networks of temporary accounts. To further break the traceable link between sender and receiver, illicit actors frequently use cyclic transfers and cryptocurrency **mixers** services that pool together funds from multiple users, mix them, and redistribute them at random intervals and amounts. These advanced tactics create dense, highly camouflaged transaction topologies that easily evade threshold-based rules and simple heuristic detection methods.

To achive their goal they utilize a strategy of transaction layering, often characterized by a topological depth L, and network diversification to obscure the money trail [5, 9]. By spreading the funds through a complex topology, the fraud ring successfully increases its non-detection probability p_L, which now depends explicitly on the layering depth L. The expected profit of a fraud ring utilizing L layers can be mathematically expressed as:

$$E[\pi_L] = p_L x - (1 - p_L)T - c(L) \tag{2.2}$$

where $E[\pi_L]$ is the expected profit, p_L is the enhanced probability of evading detection when L layers are used, x is the total illicit funds, T is the total penalty if caught, and $c(L)$ represents the monotonic cost function associated with establishing and maintaining the L transaction layers. Because the fraud ring dramatically increases the probability of evading detection ($p_L \to 1$ as L increases), the expected payoff remains highly profitable even when factoring in the complex layering costs $c(L)$ [5].

2.3 ML PATTERNS

3

ETHICS, PRIVACY AND REGULATIONS

4

STATE OF THE ART

5

PROPOSED APPROACH AND METHODOLOGY

6

EXPERIMENTS

6.1 experimental settings

6.2 hyperparameter tuning

6.3 experimental results and discussion

7

CHALLENGES

8

CONCLUSION

9

FUTURE DIRECTIONS

A

SOFTWARE

BIBLIOGRAPHY

[1] Canadian Anti-Fraud Centre. Canadian anti-fraud centre. `https://antifraudcentre-centreantifraude.ca/index-eng.htm`, 2026. Accessed: 2026-08-17. (Cited on page 7.)

[2] Ana Isabel Canhoto. Leveraging machine learning in the global fight against money laundering and terrorism financing: An affordances perspective. *Journal of Business Research*, 131:441–452, 2021. (Cited on page 8.)

[3] Dawei Cheng, Yujia Ye, Sheng Xiang, Zhenwei Ma, Ying Zhang, and Changjun Jiang. Anti-money laundering by group-aware deep graph learning. *IEEE Transactions on Knowledge and Data Engineering*, 35(12):12444–12457, 2023. (Cited on page 8.)

[4] Dawei Cheng, Yao Zou, Sheng Xiang, and Changjun Jiang. Graph neural networks for financial fraud detection: a review. *Frontiers of Computer Science*, 19(9), January 2025. (Cited on page 9.)

[5] Alfredo Contreras and Edgar Villa Pérez. Strategic choices in money laundering: Smurfing, layering, and financial over-diversification. *Journal of Economic Criminology*, 11:100199, 2026. (Cited on pages 11 and 12.)

[6] Mariano-Florentino Cuéllar. The tenuous relationship between the knowing act of money laundering and its underlying crimes. *Columbia Law Review*, 103(2):312–431, 2003. (Cited on page 9.)

[7] Europol. Does crime still pay? criminal asset recovery in the eu. Technical report, Europol, 2016. (Cited on page 7.)

[8] Ouren Kuiper, Martin van den Berg, Joost van der Burgt, and Stefan Leijnen. Exploring explainable ai in the financial sector: Perspectives of banks and supervisory authorities. In *Benelux Conference on Artificial Intelligence*, pages 253–268. Springer, 2021. (Cited on page 8.)

[9] Xiaoqi Li, Wenkai Li, Zhijie Liu, Meikang Qiu, Zhiquan Liu, Sen Nie, Zongwei Li, Shi Wu, and Yuqing Zhang. Scamsweeper: Detecting illegal accounts in web3 scams via transactions analysis. *IEEE Transactions on Information Forensics and Security*, 2025. (Cited on page 12.)

[10] Adetoyese Omoseebi, Godwin Ola, and Jackson Tyler. Rule-based systems in aml. 03 2025. (Cited on pages 7 and 8.)

[11] K. Singh and P. Best. Anti-money laundering: Using data visualization to identify suspicious activity. *International Journal of Accounting Information Systems*, 34:100418, 2019. (Cited on page 9.)

[12] United Nations Office on Drugs and Crime (UNODC). *Estimating illicit financial flows resulting from drug trafficking and other transnational organized crimes*. UNODC Vienna, 2011. (Cited on page 7.)